

Online Examination Management System



Submitted By

Abdul Basit

Enrolment No.221428

Supervised By:

Muhammad Fayaz

Lecturer

Edwardes College the Mall Peshawar Cantt.

affiliated with

University of Peshawar

2022—2026

DECLARATION

I hereby declare that the work described in this dissertation was carried out by me under the supervision of Prof. Abdul Hamid at the Department of Computer Science, Edwardes College of Peshawar, Pakistan.

I also hereby declare that the substance of this dissertation has neither been submitted elsewhere nor is being concurrently submitted for any other degree.

I further declare that the dissertation embodies the results of my own research or advance studies and that it has been composed by me and where appropriate, I have made acknowledgement to the work of others.

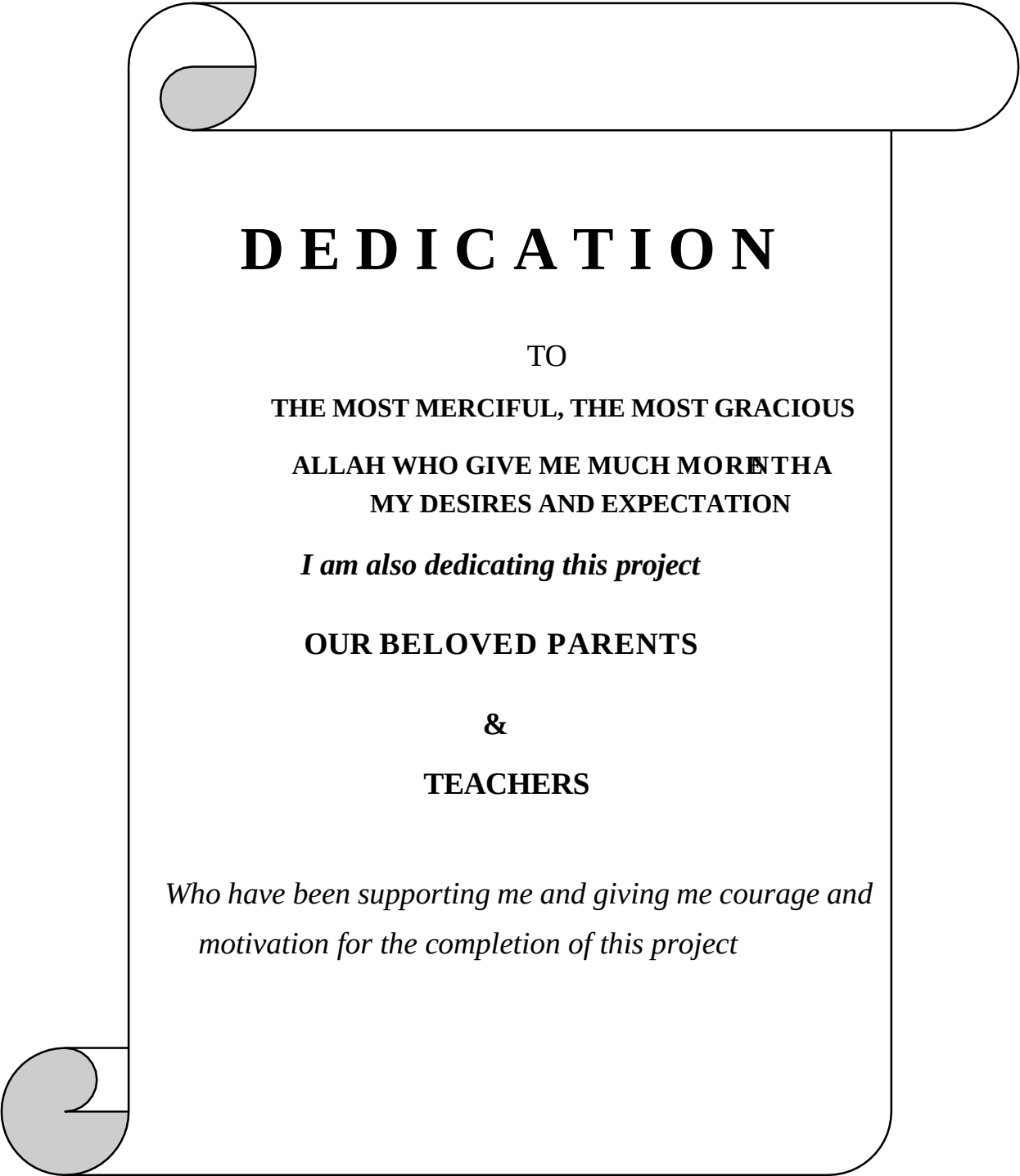
Abdul Basit

BS Computer Science
Enrolment No. 221428

CERTIFICATE

I certify that I have ensured that Mr. Abdul Basit has complied with the regulations for higher degrees. I also certify that the research work embodied in this dissertation entitled “*Online Examination Management System*” under supervision of Prof. Abdul Hameed and is worth presenting to University of Peshawar, Pakistan for the award of the degree for *BS Computer Science*.

(*Supervisor*)



D E D I C A T I O N

TO

THE MOST MERCIFUL, THE MOST GRACIOUS

**ALLAH WHO GIVE ME MUCH MORE THAN
MY DESIRES AND EXPECTATION**

I am also dedicating this project

OUR BELOVED PARENTS

&

TEACHERS

*Who have been supporting me and giving me courage and
motivation for the completion of this project*

(2022—2026)

Abstract

OEMS aims to transform traditional exam processes by providing a secure, automated, and scalable platform for online assessments. This Work outlines the design, implementation and validation of a full web-based Examination Management System A secure, high-performance web platform that will revolutionize the traditional academic testing by becoming the digital form of assessment. The system was made on the basis of three tier php/MySQL architecture and is based on a hierarchical model with structured level Mega Examinations structured with subject-specific Papers with mixed question formats, and objective grading and manual subjective evaluation.

The platform follows rigid role-based processes: Administrators manage institutional evaluation processes and allow examination content Teacher's author, administer and score paper, with adjustable parameters Students take assessment tests with time limits and get back feedback with performance analytics. Formal approval system would ensure a training of the academics; automated calculation of the percentage would ensure automatic results and institution reporting would provide a sense of transparency and efficiency. The system has been proven through complete functional validation to provide a robust solution that improves upon the integrity of examinations, decreases administrative demands and provide a scalable framework on which modern day digital assessment is achievable within academic settings.

TABLE OF CONTENTS

LIST OF FIGURES	10
CHAPTER 1: FINAL PROJECT PROPOSAL	11
1.1 INTRODUCTION.....	12
1.2 PROJECT TITLE.....	12
1.3 PROJECT OVERVIEW STATEMENT.....	12
1.4 PROJECT GOALS & OBJECTIVES	13
1.5 HIGH-LEVEL SYSTEM COMPONENTS.....	13
1.6 LIST OF OPTIONAL FUNCTIONAL UNITS	13
1.7 EXCLUSIONS.....	14
1.8 APPLICATION ARCHITECTURE	14
1.9 GANTT CHART	15
1.10 HARDWARE AND SOFTWARE SPECIFICATION	15
1.11 TOOLS AND TECHNOLOGIES USED WITH REASONING.....	16
CHAPTER 2: LITERATURE REVIEW.....	18
2.1 EVOLUTION OF DIGITAL ASSESSMENT SYSTEMS.....	19
2.2 THEORETICAL FRAMEWORKS FOR ONLINE ASSESSMENT	19
2.2.1 <i>Constructivist Assessment Theory</i>	19
2.2.2 <i>Technology Acceptance Model (TAM)</i>	19
2.2.3 <i>Assessment Security Frameworks</i>	20
2.3 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS	20
2.3.1 <i>Commercial Platforms</i>	20
2.3.2 <i>Open Source Alternatives</i>	20
2.3.3 <i>Custom Institutional Systems</i>	20
2.4 TECHNOLOGICAL FRAMEWORKS FOR E-ASSESSMENT.....	21
2.4.1 <i>Three-Tier Architecture</i>	21
2.4.2 <i>Database Design Patterns</i>	21
2.4.3 <i>Security Implementation</i>	21
2.5 PEDAGOGICAL IMPLICATIONS OF DIGITAL ASSESSMENT.....	21
2.5.1 <i>Feedback Efficacy</i>	21
2.5.2 <i>Accessibility Enhancement</i>	22
2.5.3 <i>Assessment for Learning</i>	22
2.6 SECURITY CONSIDERATIONS IN ONLINE EXAMINATIONS	22
2.6.1 <i>Academic Integrity</i>	22
2.6.2 <i>Data Privacy</i>	22
2.7 RESEARCH GAP IDENTIFICATION.....	22
2.7.1 <i>Cost-Effective Institutional Solutions</i>	23
2.7.2 <i>Contextual Adaptation</i>	23
2.7.3 <i>Balanced Automation</i>	23
2.7.4 <i>Implementation in Resource-Constrained Settings</i>	23
2.8 THEORETICAL CONTRIBUTIONS OF THIS RESEARCH	23
2.8.1 <i>Educational Technology</i>	23

2.8.2 Software Engineering Education	23
2.8.3 Human-Computer Interaction.....	24
2.8.4 Institutional Technology Adoption.....	24
CHAPTER 3: SECOND DELIVERABLE FOR OBJECT ORIENTED APPROACH	25
3.1 INTRODUCTION:	26
3.1.1 System Specifications	26
3.1.2 Identifying External Entities	27
3.1.3 Context Level Data Flow Diagram.....	27
3.1.4 Capture "Shall" Statements.....	27
3.1.5 Allocate Requirements	28
3.1.6 Prioritize Requirements	28
3.1.7 Requirements Traceability Matrix	29
3.2 HIGH-LEVEL USE CASE DIAGRAM	29
CHAPTER 4: THIRD DELIVERABLE FOR OBJECT ORIENTED APPROACH	33
4.1 INTRODUCTION.....	33
4.2 DOMAIN MODEL	34
4.3 SYSTEM SEQUENCE DIAGRAM	34
4.4 SEQUENCE DIAGRAM	35
4.5 DESIGN CLASS DIAGRAM.....	36
CHAPTER 5: 2ND & 3RD DELIVERABLE FOR STRUCTURED APPROACH.....	38
5.1 INTRODUCTION.....	38
5.2 ENTITY RELATIONSHIP DIAGRAM (ERD).....	39
5.3 DATA FLOW DIAGRAM (FUNCTIONAL MODEL).....	40
5.4 STATE TRANSITION DIAGRAM (STD).....	41
5.5 ARCHITECTURAL DESIGN.....	42
5.6 COMPONENT-LEVEL DESIGN.....	43
CHAPTER 6: 4TH DELIVERABLE (USER INTERFACE DESIGN).....	45
6.1 INTRODUCTION.....	46
6.2 SITE MAPS	46
6.3 STORYBOARDS	47
6.4 NAVIGATIONAL MAPS.....	48
6.5 TRACEABILITY MATRIX	49
CHAPTER 7: 5TH DELIVERABLE (SOFTWARE TESTING).....	51
7.1 INTRODUCTION.....	52
7.2 TEST PLAN	52
7.2.1 Purpose	52
7.2.2 Test Plan ID	52
7.2.3 Test Items	52
7.2.4 To Be Tested Features	53
7.2.5 Features Not Tested	53
7.3 TESTING STRATEGY	53
7.3.1 Pass/Fail Criteria	53
7.3.2 Suspension/Resumption Criteria.....	54
7.3.3 Deliverables	54
7.4 TESTING TASKS.....	54
7.4.1 Test Environment	54

7.4.2 Responsibilities	54
7.4.3 Training Needs.....	55
7.4.4 Schedule	55
7.5 RISKS	55
7.6 APPROVALS.....	55
APPENDIXES:	57

List of Figures

Figure 1. 1 System Workflow	17
Figure 3. 1 Context Level Diagram	27
Figure 4. 1 Domain Model.....	34
Figure 4. 2 System Sequence Diagram	35
Figure 4. 3 Sequence Diagram.....	36
Figure 4. 4 Class Diagram	37
Figure 5. 1 ER Diagram	40
Figure 5. 2 Data Flow Diagram	41
Figure 5. 3 State Transition Diagram.....	42
Figure 5. 4 Architectural Design.....	43
Figure 5. 5 Component Level Diagram	44
Figure 8. 1 Admin Dashedboard.....	57
Figure 8. 2 Mega Exam panel	58
Figure 8. 3 Student Dashboard.....	58
Figure 8. 4 Paper Approval	59
Figure 8. 5 Student Login	59
Figure 8. 6 Question Display	60
Figure 8. 7 Paper Submission	60
Figure 8. 8 Available Exams Screen.....	60
Figure 8. 9 Evaluation Answer	61
Figure 8. 10 Results Checking.....	61
Figure 8. 11 Login Page.....	62
Figure 8. 12 Student Login	62

Chapter 1: Final Project Proposal

1.1 Introduction

With the rise of digital transformation, traditional examination methods have become inefficient, insecure, and labor-intensive. Online Examination Management System (OEMS) provides highly secure, scalable and fully automated system that helps the educational institutions to effectively carry out online tests. OEMS caters for the whole examination cycle, including the development and validation of question papers, the grading and results announcement process, in turn creating the levels of accessibility and ethical standards of academic practice. Global changes like the Covid-19 pandemic have increased this digital transformation as it has brought a heavy realization about how crucial strong remote assessment solutions are which can provide examination standards without having an influence on the security or fairness. The system fixes the problems of paper-based examination system by ensuring that they provide a full digital solution which will save on administrative burden and enhance the exam experience for all the stakeholders.

1.2 Project Title

Online Examination Management System (OEMS)

1.3 Project Overview Statement

OEMS is a web-based platform designed to modernize the examination process by automating exam creation, distribution, monitoring, and result processing. It supports remote proctoring, role-based access, multiple question formats, and real-time result generation. The system addresses the limitations of paper-based exams and reduces administrative burdens, providing scalability and inclusiveness for students, instructors, and administrators. The platform facilitates a seamless experience in examining with various checks and balances in place to keep students honest and passage like everyone else. It helps in conducting examinations efficiently, safely and more accessible for all who participate.

1.4 Project Goals & Objectives

The primary goals of OEMS are to improve the efficiency, security and accessibility of assessments. Specific objectives include:

- Automate exam creation, grading, and result reporting.
- Provide AI-based proctoring and multi-layered security.
- Enable remote access with role-based user management.
- Support a wide variety of question types (MCQs, essays, coding tasks).
- Offer assistive features for students with special needs.
- Generate instant feedback and detailed performance analytics.
- Reduce operational and administrative costs.

1.5 High-Level System Components

1. Frontend Interface:

HTML, CSS, JavaScript to develop a responsive, user-friendly and user-intuitive interface at all user roles.

2. Backend Logic:

PHP – handles server-side logic, data flow, and user role controls.

3. Database:

MySQL – manages all data including users, questions, exams, and results.

4. Security & Proctoring:

AI-based monitoring tools, user authentication, and encrypted data storage.

5. Automated Grading Module:

Instant evaluation for objective questions and analytics reporting.

6. Examination Environment Controller: Full-screen mode enforcement and navigation restriction during exams

1.6 List of Optional Functional Units

- Live webcam proctoring and screen monitoring.

- Biometric authentication for high-security exams.
- Chat support or help desk integration for students.
- Multi-language support for international candidates.
- Adaptive assessments based on performance.
- Comprehensive exportable reports for academic audit and institutional analysis

1.7 Exclusions

- No integration with third-party LMS or ERP systems in the initial release.
- No support for subjective answer auto-evaluation (e.g., essay-type AI checking).
- Biometric features may be excluded due to cost or privacy concerns.
- Offline exam mode is not supported requires continuous internet access.
- The present implementation is not constituted by mobile application versions that are distributed to iOS and Android.

1.8 Application Architecture

OEMS has a **three-tier architecture** which guarantees separation of concerns, maintainability and scalability:

1. **Presentation Layer (Frontend)**

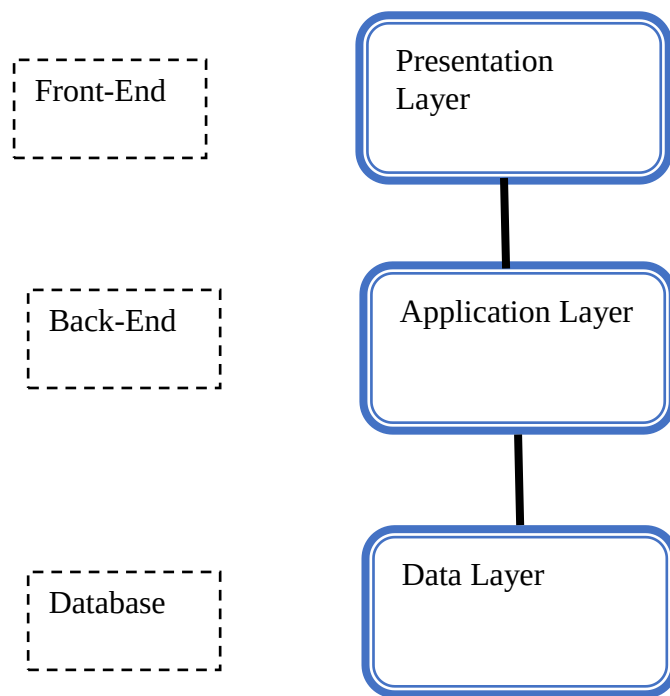
Provides an interactive and responsive user interface to students, faculty, and administration. Developed using html5, css3 and bootstrap and js/jquery

2. **Application Layer (Backend)**

PHP-based server logic handles requests with business rules, examination flow control, authentication and automated grading. This layer deals with the frontend requests and it communicates with the database layer.

3. **Data Layer (Database)**

MySQL securely stores exam data, user records, results, and logs. Data is stored in the form of normalized tables for the sake of integrity and efficient retrieval



This modular architecture ensures scalability, security, and maintainability.

1.9 Gantt Chart



1.10 Hardware and Software Specification

Hardware Requirements

- Processor: Intel Core i3 or more (in the case of server deployment)
- RAM: Minimum - 4 GB (8 is recommended for production environment)

- Storage: 10GB free disk space to setup first (can be scaled according to user base)
- Network: Stable internet connection with minimum 5Mbps bandwidth for concurrent users
- Client Devices - Standard operating computer or laptop or tablets that have web camera capabilities for proctoring functions

Software Requirements

- OS: Windows/Linux
- Web Browser: Google Chrome, Mozilla Firefox
- Development Tools: VS Code, XAMPP/WAMP
- Languages: PHP 7.4+, HTML5, CSS3, JavaScript ES6+
- Database Server: MySQL
- Web Server: Apache

1.11 Tools and Technologies Used with Reasoning

Tool / Technology	Purpose / Reason
HTML & CSS	Page layout and responsive styling
JavaScript	Enhancing interactivity and UI responsiveness
PHP	Server-side scripting and business logic
MySQL	Structured data storage for exams, users, and results
Bootstrap	Mobile-friendly UI components
XAMPP / WAMP	Local server for testing PHP and MySQL integration

The Online Examination Management System (OEMS) follows a structured methodology to ensure efficiency, security, and reliability. The development process includes the following key steps: Requirement Analysis, System Design, Implementation, Testing and Debugging, and Deployment and Maintenance. Each phase plays a crucial role in building a seamless and user-friendly online examination platform. The complete system development process is shown in Fig.1.

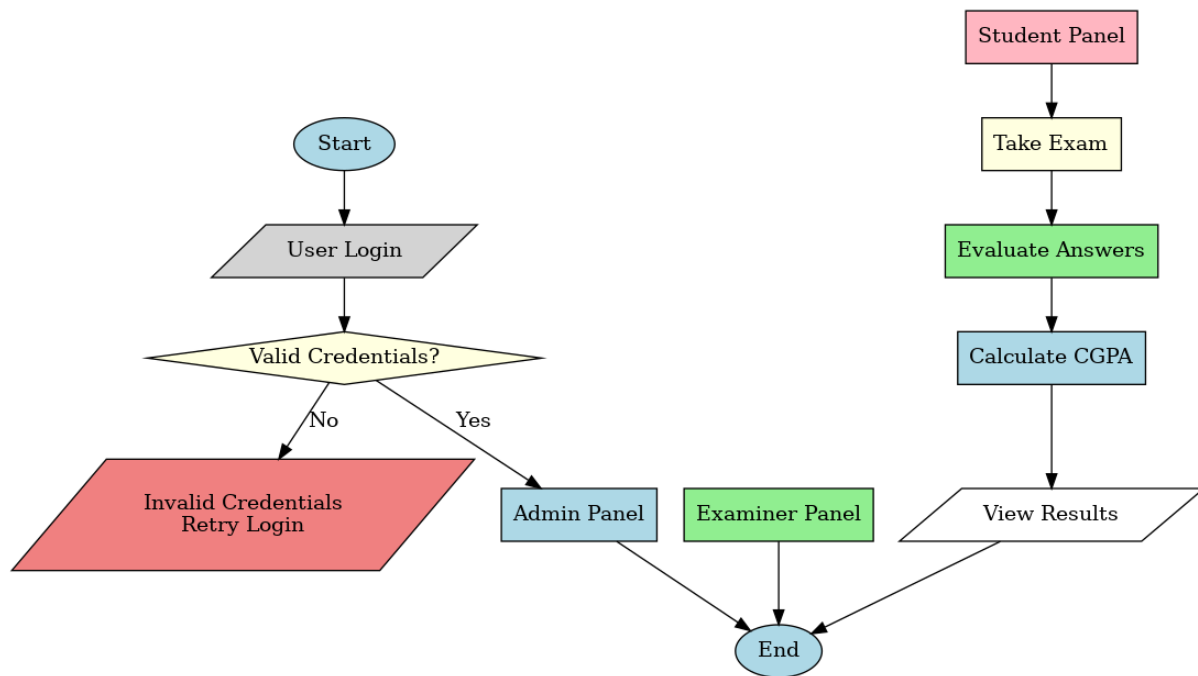


Figure 1. 1 System Workflow

CHAPTER 2: LITERATURE REVIEW

2.1 Evolution of Digital Assessment Systems

The transformation from traditional paper-based examinations to digital assessment platforms represents a significant evolution in educational technology. Early computerized testing systems emerged in the 1990s with basic multiple-choice testing capabilities, primarily used for standardized assessments like the GMAT and GRE (Bennett, 2015). These systems focused on objective question types with automated scoring but lacked comprehensive institutional workflow management.

The 2000s witnessed the integration of Learning Management Systems (LMS) with basic assessment modules. Platforms like Moodle, Blackboard, and Canvas incorporated quiz functionalities but remained limited in supporting high-stakes institutional examinations requiring security, proctoring, and complex grading schemes (Sclater, 2016).

Recent advancements, accelerated by the COVID-19 pandemic, have driven development of specialized online examination systems. Modern platforms incorporate artificial intelligence for proctoring, adaptive testing algorithms, and comprehensive analytics. However, research indicates that many current solutions remain inaccessible to institutions in developing regions due to cost constraints and infrastructure requirements (Ali et al., 2021).

2.2 Theoretical Frameworks for Online Assessment

2.2.1 Constructivist Assessment Theory

The system aligns with constructivist principles by enabling diverse assessment formats that measure not only knowledge recall but also application and synthesis. The inclusion of descriptive questions alongside multiple-choice items supports Vygotsky's Zone of Proximal Development by allowing students to demonstrate higher-order thinking skills (Vygotsky, 1978).

2.2.2 Technology Acceptance Model (TAM)

Davis's (1989) Technology Acceptance Model provides a theoretical framework for understanding user adoption of the system. Perceived usefulness and ease of use, key determinants in TAM, were prioritized in system design through intuitive interfaces and clear workflow structures for all user roles.

2.2.3 Assessment Security Frameworks

The system implements principles from institutional assessment security frameworks, including role-based access control, examination password protection, and temporal restrictions. These measures address cheating prevention strategies identified by Lancaster and Clarke (2016) while balancing usability concerns.

2.3 Comparative Analysis of Existing Systems

2.3.1 Commercial Platforms

Commercial solutions like ExamSoft, ProctorU, and Respondus offer comprehensive features but present significant barriers:

- **Cost:** Licensing fees range from \$15-50 per student annually, prohibitive for many institutions.
- **Infrastructure:** Often require specialized hardware and high-speed internet.
- **Customization:** Limited adaptability to institutional-specific workflows

2.3.2 Open-Source Alternatives

Open-source platforms like TCExam and ILIAS provide cost-effective alternatives but present challenges:

- **Technical Complexity:** Require specialized IT expertise for deployment and maintenance
- **Feature Limitations:** Often lack comprehensive institutional workflows
- **Support:** Limited formal support structures

2.3.3 Custom Institutional Systems

Custom-built systems, while tailored to specific needs, typically involve:

- **Development Costs:** RS 20,000-100,000 for initial development
- **Maintenance Burden:** Ongoing technical support requirements
- **Scalability Issues:** Often limited to specific institutional contexts

2.4 Technological Frameworks for E-Assessment

2.4.1 Three-Tier Architecture

The system adopts the three-tier web application architecture (presentation, application, data layers) widely recognized as optimal for scalable educational platforms (Sommerville, 2016). This separation of concerns facilitates maintenance, security implementation, and future enhancements.

2.4.2 Database Design Patterns

The hierarchical examination structure follows database normalization principles to minimize redundancy while maintaining referential integrity. The Mega Exam-Paper-Section-Question hierarchy represents an implementation of the composite design pattern, enabling flexible examination organization (Gamma et al., 1994).

2.4.3 Security Implementation

The system incorporates security measures based on OWASP (Open Web Application Security Project) guidelines:

- **Authentication:** Secure password hashing using bcrypt algorithm
- **Authorization:** Role-based access control (RBAC) implementation
- **Session Management:** Secure session handling with timeout mechanisms
- **Input Validation:** Server-side validation to prevent injection attacks

2.5 Pedagogical Implications of Digital Assessment

Research indicates several pedagogical advantages of well-designed online examination systems:

2.5.1 Feedback Efficacy

Digital systems enable immediate feedback for objective questions, which Hattie and Timperley (2007) identify as crucial for learning enhancement. The system's instant MCQ grading aligns

with this principle, while manual descriptive evaluation allows for nuanced, constructive feedback.

2.5.2 Accessibility Enhancement

Online systems improve accessibility for students with disabilities through adjustable interfaces and flexible scheduling options. The system's time management features accommodate diverse learner needs while maintaining assessment integrity (Burgstahler, 2015).

2.5.3 Assessment for Learning

The comprehensive analytics provided by the system support "assessment for learning" approaches, enabling data-informed pedagogical adjustments. Performance patterns across cohorts can inform curriculum refinement and targeted interventions (Black & Wiliam, 2009).

2.6 Security Considerations in Online Examinations

2.6.1 Academic Integrity

Research identifies multiple threats to academic integrity in online environments:

- **Impersonation:** Addressed through secure authentication and session management
- **Unauthorized Assistance:** Mitigated through time limits and fullscreen enforcement
- **Question Leakage:** Prevented through administrative approval workflows and access controls

2.6.2 Data Privacy

The system implements data protection measures aligned with General Data Protection Regulation (GDPR) principles, including secure data storage, access logging, and user consent for data processing.

2.7 Research Gap Identification

Despite extensive literature on online assessment, several gaps persist in implementation contexts similar to this project:

2.7.1 Cost-Effective Institutional Solutions

Limited research addresses the development of comprehensive yet affordable systems for institutions with constrained budgets. Most studies focus either on commercial platforms or basic open-source solutions without institutional workflow integration.

2.7.2 Contextual Adaptation

Few systems demonstrate successful adaptation to specific institutional contexts with unique examination structures and approval processes. The Mega Exam hierarchy represents an innovation in this regard.

2.7.3 Balanced Automation

Existing literature tends to focus on either fully automated systems or entirely manual processes. This system's hybrid approach automating objective components while retaining human judgment for subjective evaluation represents an underexplored middle ground.

2.7.4 Implementation in Resource-Constrained Settings

There is limited documentation of successful implementation and evaluation of online examination systems in institutions with basic technological infrastructure, a gap this research addresses.

2.8 Theoretical Contributions of This Research

This project contributes to several theoretical domains:

2.8.1 Educational Technology

Provides a case study in implementing hierarchical assessment structures within digital platforms, demonstrating how institutional examination workflows can be translated to online environments while maintaining academic standards.

2.8.2 Software Engineering Education

Offers insights into developing complex educational systems using standard web technologies, providing a reference implementation for similar projects.

2.8.3 Human-Computer Interaction

Demonstrates interface design principles for multi-role educational systems, balancing functionality with usability across diverse user groups with varying technical proficiency.

2.8.4 Institutional Technology Adoption

Contributes to understanding factors influencing technology adoption in educational institutions, particularly regarding assessment systems where security and reliability concerns often impede digital transformation.

Chapter 3: Second Deliverable for Object Oriented Approach

3.1 Introduction:

Requirements engineering process provides the correct mechanism for comprehension of what the customer desires, evaluating needs, determining feasibility, negotiating a fair solution, describing the solution in an unambiguous manner, verifying the specification, and dealing with the requirements as they are translated into an operational system. The work entails capturing, organizing, and actually depicting the requirements of the user so that they can be properly embodied in systems which fulfill those requirements (i.e., good quality).

- Requirement's elicitation
- Requirements analysis and negotiation
- Requirements specification
- System modeling
- Requirements validation
- Requirements management

3.1.1 System Specifications

The Student Management System with Percentage Calculation is a centralized, web-based application providing secure management of student profiles, course registrations, result entry, and computerized Percentage calculation. It excludes manual processing, minimizes errors, and offers students instant access to academic performance.

Existing System:

Nowadays, student academic records, grades and Percentage calculation are done manually or by patchy systems. This leads to inconsistency of data, errors in calculations, report generation delays, and restricted access to students.

Scope of the System:

The planned system will offer:

1. Secure student registration and profile management

2. Faculty portal for course-wise marks entry
3. Real-time Percentage calculation from entered grades
4. Academic transcript generation on-demand
5. Role-based access for students, faculty, and administrators
6. Reports for monitoring student performance

3.1.2 Identifying External Entities

Entity	Description
Student	Can register, view courses, check Percentage and transcript.
Faculty	Manages course records, uploads marks.
Admin	Manages users, course offerings, and system control.

3.1.3 Context Level Data Flow Diagram

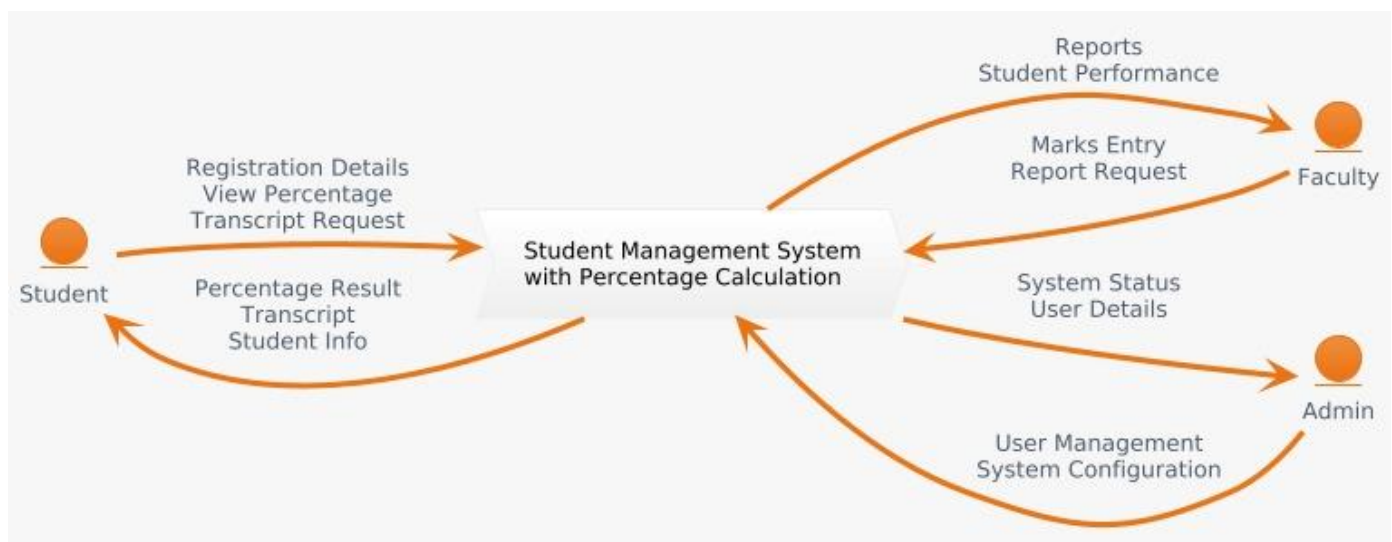


Figure 3. 1 Context Level Diagram

3.1.4 Capture "Shall" Statements

ID	Requirement Statement
R1	The system shall allow students to register and login.
R2	The system shall allow students to view academic records and Percentage.
R3	The system shall allow faculty to add marks and manage course results.
R4	The system shall calculate Percentage based on entered grades automatically.
R5	The system shall generate transcripts for students.
R6	The system shall allow admin to manage users, courses, and system configurations.
R7	The system shall provide secure authentication for all roles.

3.1.5 Allocate Requirements

Requirement	Use Case ID	Use Case Name
Student Registration & Login	UC01	UC_StudentAuth
View Academic Records & Percentage	UC02	UC_ViewPercentage
Faculty Adds Marks	UC03	UC_FacultyAddMarks
System Calculates Percentage Automatically	UC04	UC_CalculatePercentage
Generate Transcripts	UC05	UC_GenerateTranscript
Admin Manages System	UC06	UC_AdminManagement
Secure Authentication for All Roles	UC07	UC_SecureAccess

3.1.6 Prioritize Requirements

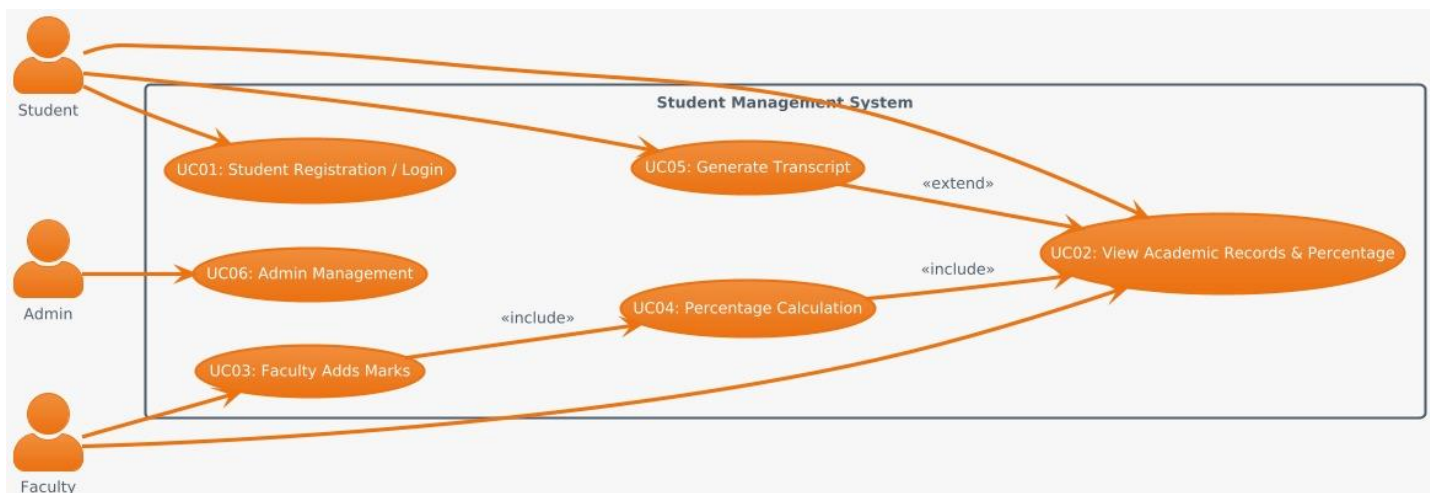
Priority	Use Case ID	Use Case Name	Summary
High	UC01	UC_StudentAuth	Login/Registration essential for access.
High	UC02	UC_ViewPercentage	Viewing Percentage is core student functionality.
High	UC04	UC_CalculatePercentage	Percentage calculation is primary system goal.
Medium	UC03	UC_FacultyAddMarks	Faculty updates grades post evaluation.

Medium	UC05	UC_GenerateTranscript	Academic transcripts on demand.
Medium	UC06	UC_AdminManagement	Admin configurations maintain the system.
High	UC07	UC_SecureAccess	Secure access mandatory for all users.

3.1.7 Requirements Traceability Matrix

Requirement	Use Case ID	Use Case Name	Priority
Student Registration & Login	UC01	UC_StudentAuth	High
View Academic Records & Percentage	UC02	UC_ViewPercentage	High
Faculty Adds Marks	UC03	UC_FacultyAddMarks	Medium
Percentage Calculation	UC04	UC_CalculatePercentage	High
Transcript Generation	UC05	UC_GenerateTranscript	Medium
Admin System Management	UC06	UC_AdminManagement	Medium
Secure Access Control	UC07	UC_SecureAccess	High

3.2 High-Level Use Case Diagram



3.2.1 Use Case Descriptions

UC01: Student Registration/Login

Attribute	Details
Use Case ID	UC01
Use Case Name	Student Registration/Login
Actor	Student
Brief Description	Allows students to create an account and login securely.
Preconditions	Student must have valid university ID and email.
Basic Flow	1. Open registration/login page 2. Enter credentials 3. System validates 4. Account created or access granted
Alternate Flows	- Email already registered → Show error - Invalid details → Reject request
Postconditions	Student account created or access granted.

UC02: View Academic Records & Percentage

Attribute	Details
Use Case ID	UC02
Use Case Name	View Academic Records & Percentage
Actor	Student, Faculty
Brief Description	Allows viewing of courses, grades, and calculated Percentage.
Preconditions	User must be logged in.
Basic Flow	1. Access dashboard 2. View academic records and Percentage
Alternate Flows	- System error → Display error message
Postconditions	Academic records displayed successfully.

UC03: Faculty Adds Marks

Attribute	Details
Use Case ID	UC03
Use Case Name	Faculty Adds Marks

Actor	Faculty
Brief Description	Faculty can enter marks for enrolled students.
Preconditions	Faculty must be logged in.
Basic Flow	1. Select course 2. Enter student marks 3. Submit marks
Alternate Flows	- Invalid marks entered → Show error message
Postconditions	Marks saved successfully; Percentage recalculated.

UC04: Percentage Calculation

Attribute	Details
Use Case ID	UC04
Use Case Name	Percentage Calculation
Actor	System (Automated)
Brief Description	System automatically calculates Percentage based on grades.
Preconditions	Marks must be available for all relevant courses.
Basic Flow	1. Fetch grades 2. Apply Percentage formula 3. Update student profile
Alternate Flows	- Missing grade data → Skip calculation
Postconditions	Percentage updated in student records.

UC05: Generate Transcript

Attribute	Details
Use Case ID	UC05
Use Case Name	Generate Transcript
Actor	Student
Brief Description	System generates official transcript in PDF format.
Preconditions	Student must be logged in; results must be available.
Basic Flow	1. Request transcript 2. System compiles data 3. PDF generated and

	provided
Alternate Flows	- No results available → Show error message
Postconditions	Transcript generated and made available to student.

UC06: Admin Management

Attribute	Details
Use Case ID	UC06
Use Case Name	Admin Management
Actor	Admin
Brief Description	Admin manages user accounts, course offerings, and settings.
Preconditions	Must be logged in as Admin.
Basic Flow	1. Access admin dashboard 2. Add/edit users 3. Configure courses and system settings
Alternate Flows	- Invalid input → Display error message
Postconditions	System records updated successfully.

Chapter 4: Third Deliverable for Object Oriented Approach

4.1 Introduction

This chapter shows the Object-Oriented Design (OOD) of the Student Management System with Percentage Calculation. Object-oriented analysis is concerned with discovering dominant system entities, their attributes, operations, and their interactions. Following diagrams are presented:

- ✓ Domain Model
- ✓ System Sequence Diagram
- ✓ Sequence Diagram
- ✓ Design Class Diagram

4.2 Domain Model

The Domain Model graphically depicts entities of the real world (objects) in the system and their interconnections. It is concerned with finding classes, their properties, and relationships that exist prior to coding

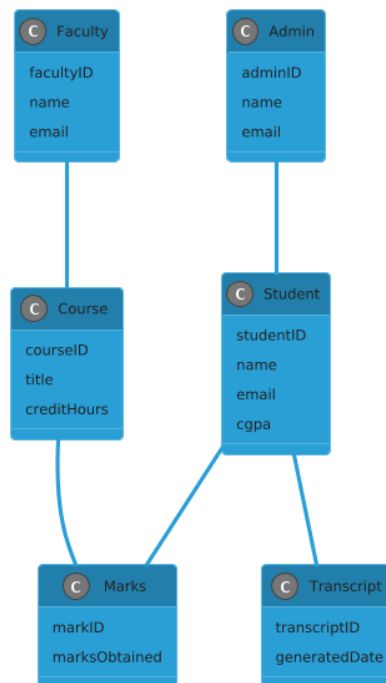


Figure 4.1 Domain Model

4.3 System Sequence Diagram

A System Sequence Diagram (SSD) illustrates interactions of external parties (students, staff, and admin) with the system for critical functionalities. It emphasizes the sequencing of messages exchanged.

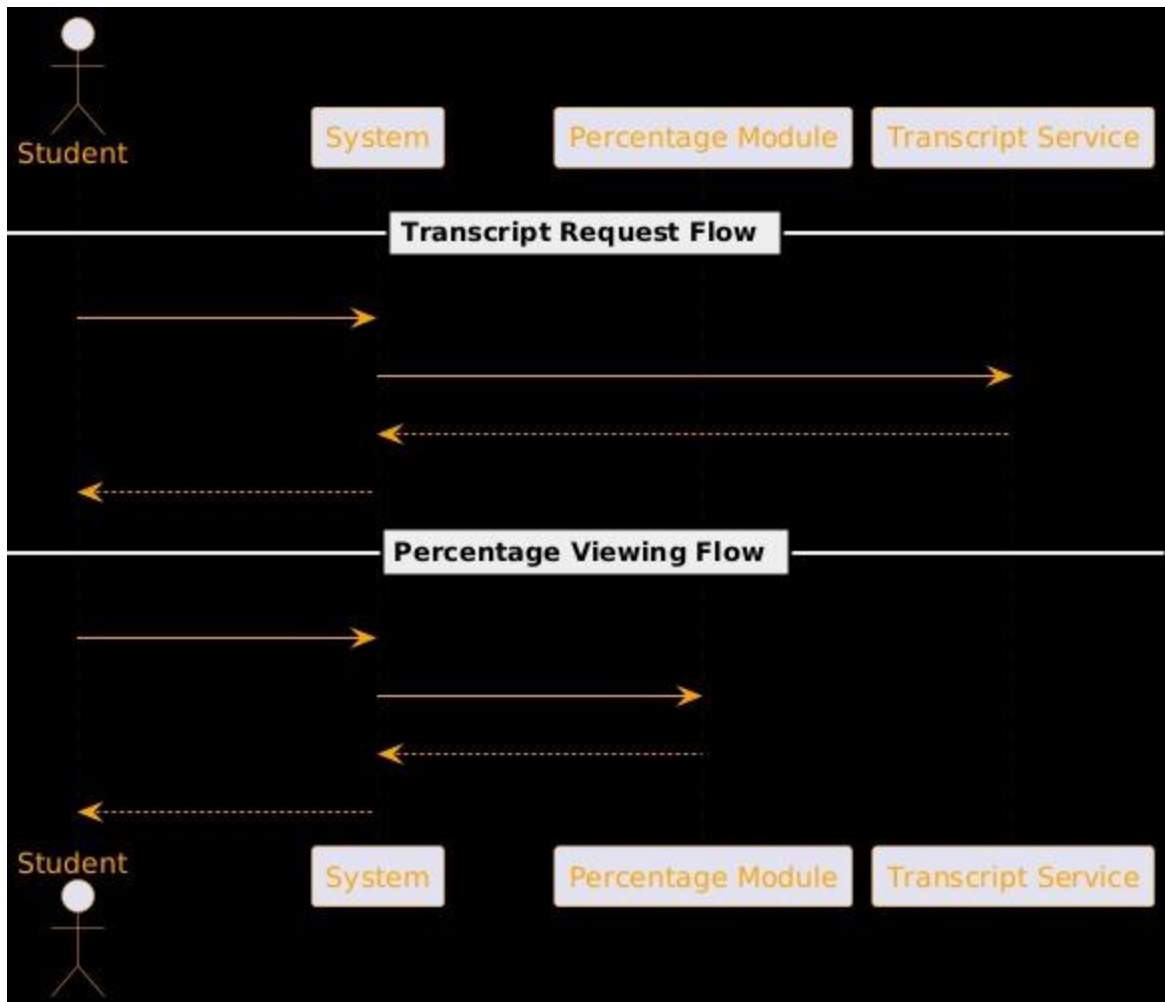


Figure 4. 2 System Sequence Diagram

4.4 Sequence Diagram

The Sequence Diagram shows a step-by-step, detailed interaction among system elements during certain operations, with the focus on method calls and message passing.

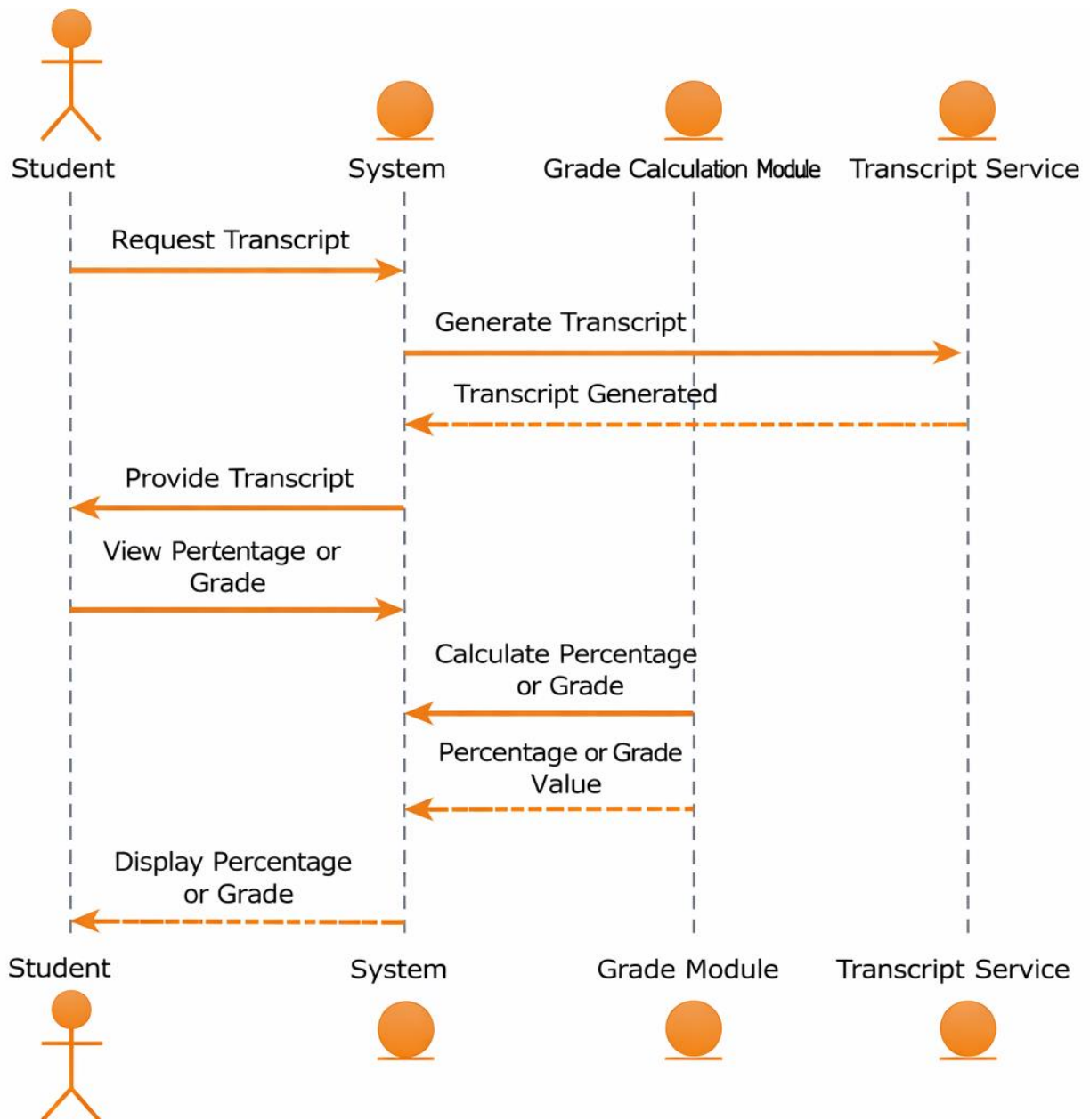


Figure 4. 3 Sequence Diagram

4.5 Design Class Diagram

The Design Class Diagram determines the class structure, its properties, operations (methods), and how they relate to each other. It informs actual implementation in code.

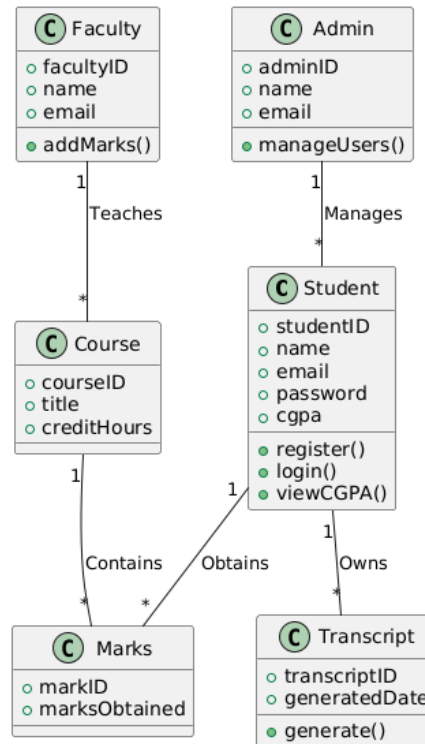


Figure 4. 4 Class Diagram

Chapter 5: 2nd & 3rd Deliverable For structured Approach

5.1 Introduction

A systematic system analysis and design process offers the basis for effectively comprehending, modeling, and realizing software solutions. This systematic process enables the stakeholders and the software engineers to see the functional and behavioral aspects of the system prior to starting the development. The Analysis and Design Models lead the project team with no uncertainty,

removing ambiguity, making things clearer, and delivering a technical roadmap. During the design and analysis stage of Student Management System with Percentage Calculation, a number of key artifacts have been developed to facilitate structured development:

5.2 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram establishes the static data structure of the system. It identifies primary entities, their attributes, and how they relate. ERDs are necessary for designing effective, normalized databases and for ensuring student academic records, grades, and calculations for Percentage are well organized.

For the Student Management System, main entities are:

- Student
- Faculty
- Admin
- Course
- Marks
- Transcript

These entities are related to each other to facilitate smooth data handling and aid in automated Percentage calculation.

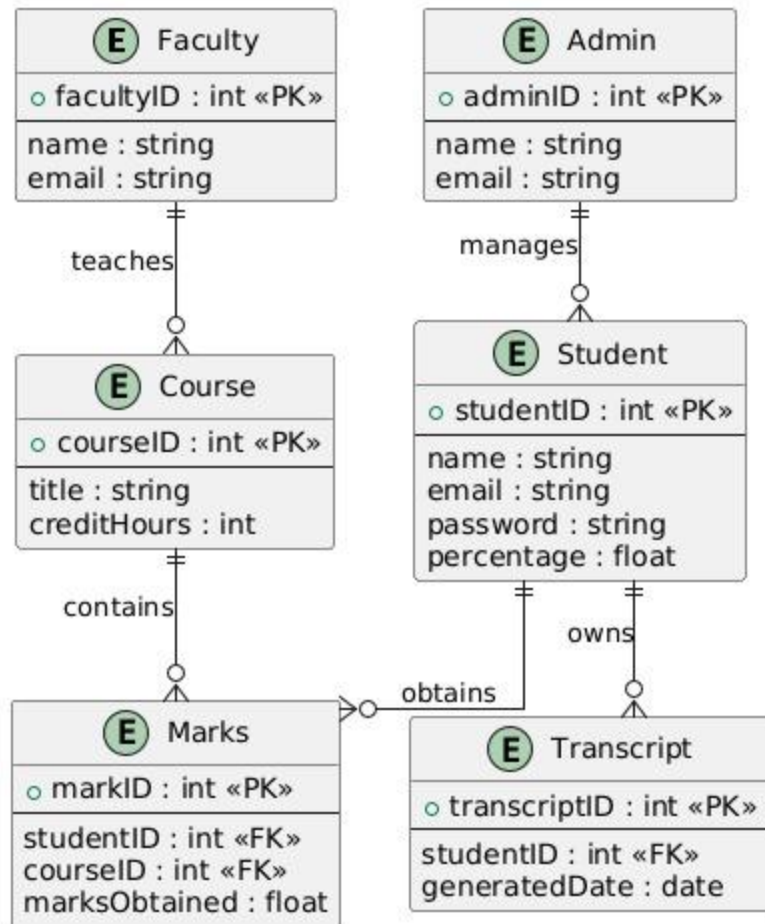


Figure 5. 1 ER Diagram

5.3 Data Flow Diagram (Functional Model)

Data Flow Diagrams (DFDs) give the functional representation of the system. They show how data flows around the system and how input is converted into output via various processes.

DFDs are developed at multiple levels to continually build system detail:

- ✓ Context Level DFD – Gives a high-level picture of system boundaries and external entities.
- ✓ Level 1 DFD – Divides the system into big processes, illustrating internal data flows.

These diagrams assist in picturing the structure of the system, with the intent to make it easy for developers, testers, and stakeholders.

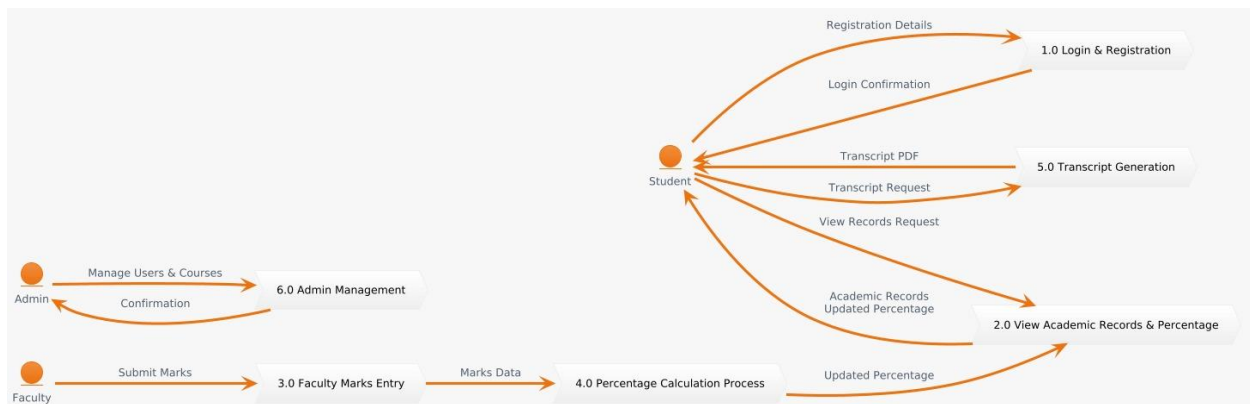


Figure 5. 2 Data Flow Diagram

5.4 State Transition Diagram (STD)

The State Transition Diagram represents the dynamic behavior of the system through the illustration of how the system transitions from one state to another based on events.

For instance, in the Student Management System:

- A student is moved from the "Not Logged In" state to the "Logged In" state after a login.
- A marks entry process is moved from "Pending" to "Completed" when submitted by faculty.

STDs assist in defining system control logic and time-dependent behavior and prevent state management errors at development time.

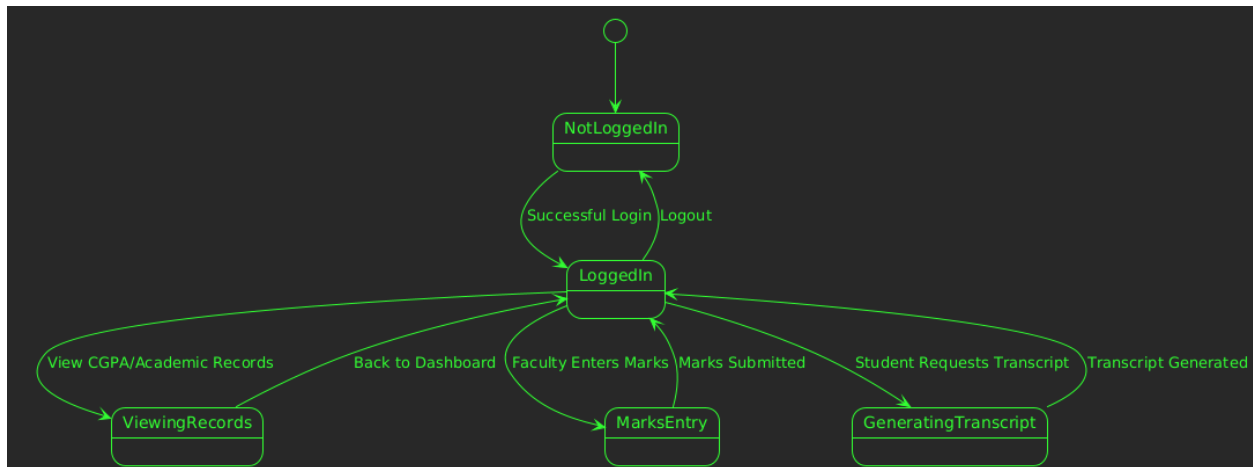


Figure 5. 3 State Transition Diagram

5.5 Architectural Design

The architectural design maps system requirements into an initial high-level structure of the software. It establishes the principal components, their interactions, and communication flow.

- For Student Management System, the architecture is of layered type:
- Presentation Layer: Student, faculty, and admin user interfaces
- Business Logic Layer: Provides Percentage calculation, transcript generation, user management
- Data Access Layer: Deals with communication to the system database

This architecture is modifiable, scalable, and easy to maintain in the future.

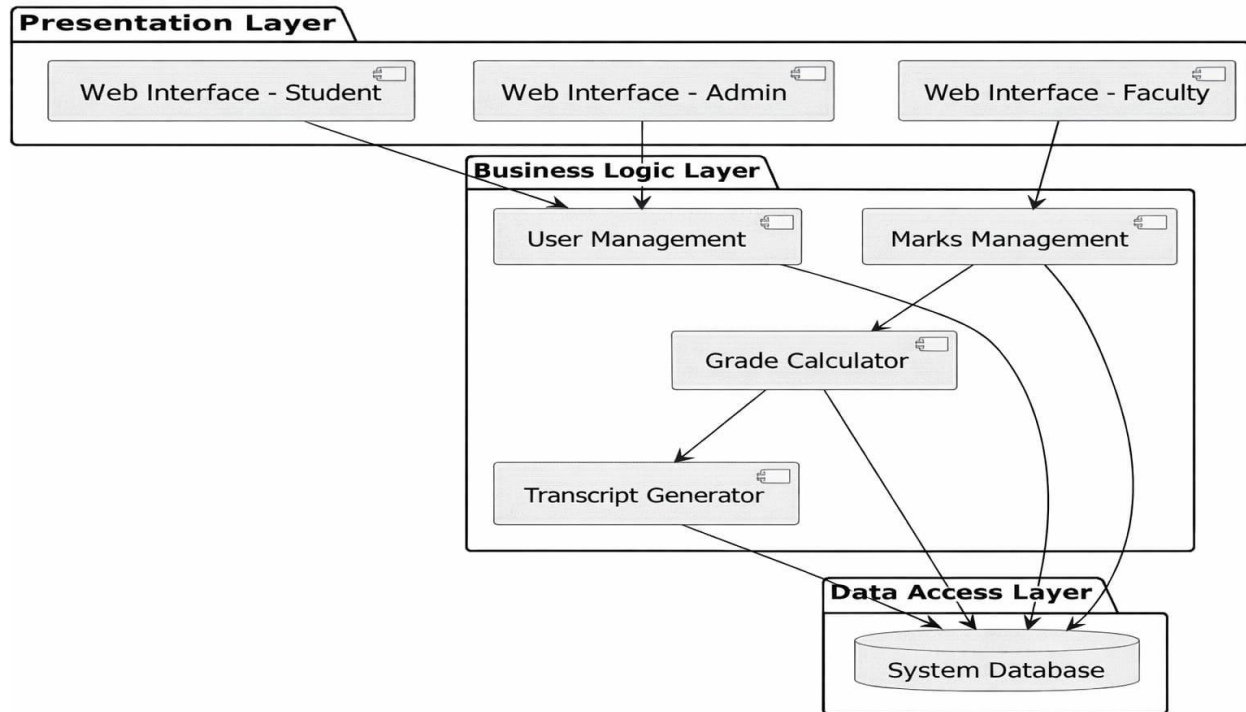


Figure 5. 4 Architectural Design

5.6 Component-Level Design

Component-level design gives a clear picture of system components or modules specified in the architecture. Internal structure of each component is described with diagrams, pseudocode, or flowcharts.

For this project, following components like:

- Percentage Calculation Module
- Transcript Generation Module
- User Authentication Service
- Marks Management System

.are designed with certain logic and internal workflows to facilitate proper, efficient development and coding.

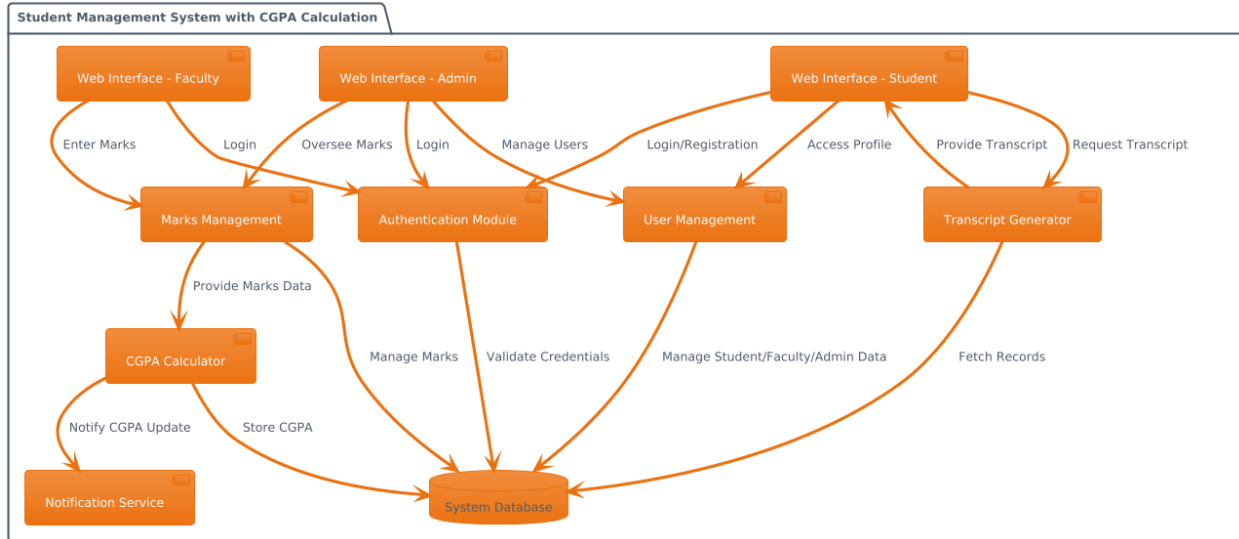


Figure 5. 5 Component Level Diagram

Chapter 6: 4th Deliverable (User Interface Design)

6.1 Introduction

For the Percentage Calculation in the Student Management System, a user-friendly and accessible User Interface (UI) is essential to help students, faculty, and administrative staff easily deal with academic records, check Percentage, and execute system operations. The design of UI incorporates four essential planning elements prior to initiation of development:

- Site Maps – defining the overall layout and screen hierarchy of the system
- Storyboards – visually representing user interaction sequences
- Navigation Maps – showing screen-to-screen navigation in accordance with user actions
- Traceability Matrix – relating system functionality to use cases, interfaces, and database components

Pre-development visualization, e.g., sketches and interaction diagrams, aids effective, user-focused design. Effective planning guarantees reasonable navigation, effective information retrieval, and effortless Percentage monitoring for the entire system users.

6.2 Site Maps

A site map gives a top-level view of the structure of the system, highlighting principal modules and how they relate to one another. For the Student Management System with Percentage Calculation, the site map clarifies the path of navigation for various user roles:

- Home Page
- Login/Registration (Students, Faculty, Admin)
- Student Dashboard
- View Academic Records
- Percentage Calculation and Display
- Transcript Generation
- Faculty Dashboard
- Marks Entry Module

- Admin Management Panel

The site is planned to be static and straightforward and should not include too dynamic elements like collapsible trees or animation to keep it visually clear. Quick, direct access to student and faculty records and Percentage is essential for students and faculty in a university environment.

6.3 Storyboards

Storyboards are graphical representations of how users interact with the system in various scenarios. In the case of the Student Management System, storyboards depict major interactions such as:

- A student accessing the portal and seeing their Percentage
- A professor inputting marks for students taking their courses
- An admin dealing with user accounts and course registration
- A student creating their academic transcript

Storyboard Elements are:

- Environment: University campus, computer lab, mobile phone
- Visual Cues: Login buttons, Percentage showing, Transcript requests
- Tactile Cues: Pressing or tapping buttons on desktop or mobile
- User Inputs: Inputting credentials, making selections, filling in forms
- Machine Outputs: Confirmation messages, Percentage values, generated transcripts
- User Emotions: Confidence, satisfaction in system feedback
- Technology Used: Web browsers, mobile applications, database backend
- Experience Quality: Fast access to academic records, accurate Percentage updates

The storyboard process starts with sketchy wireframes and develops into high-fidelity UI mockups, maintaining consistency in layout and user experience.

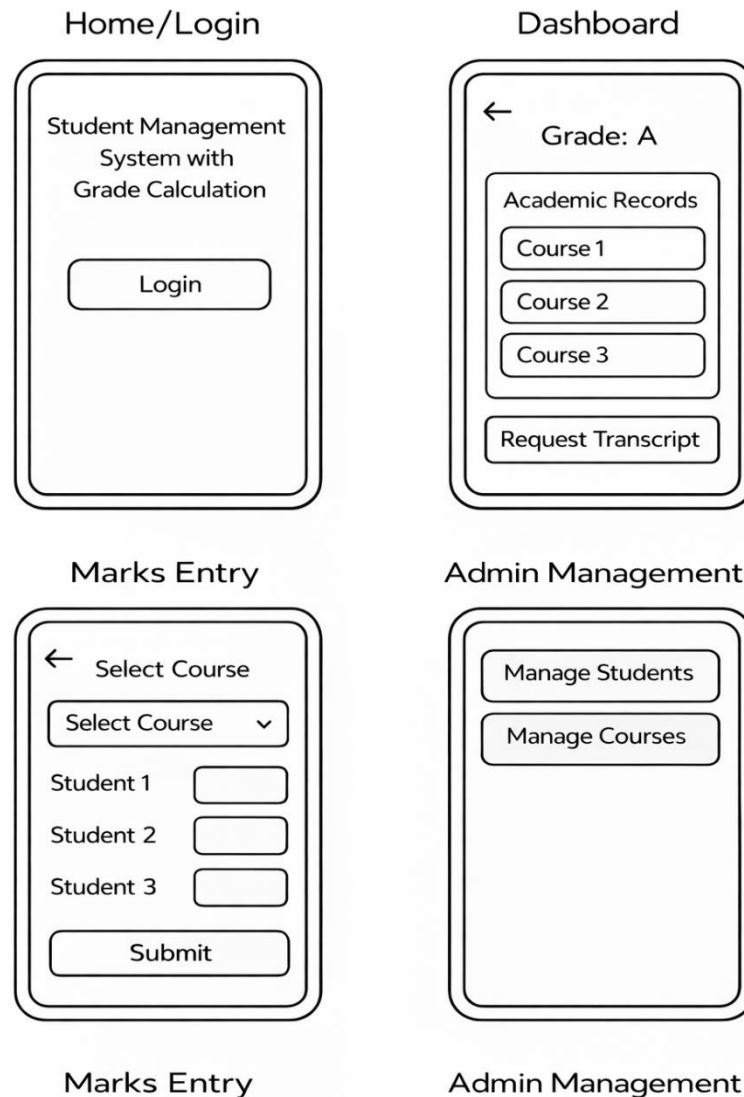


Figure 6.1 Storyboard

6.4 Navigational Maps

Navigation maps show how users move between system screens based on user actions. They depict routes for common interactions. Every arrow in the navigation map is mapped to a user

action that causes a transition of screens. Unambiguous, readable navigation keeps the system user-friendly and effective for all users.

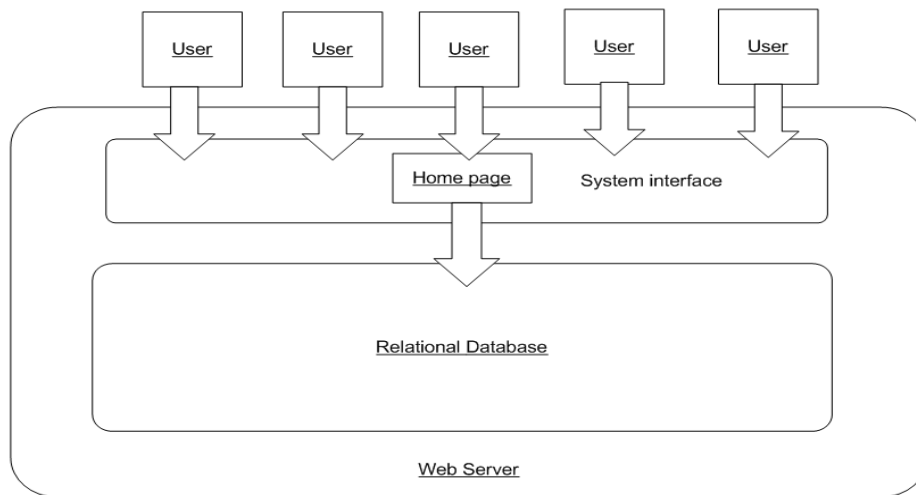


Figure 6. 2 Navigation Map

6.5 Traceability Matrix

Traceability Matrix connects system features with use cases, interface IDs, database tables, and dependent components. All functional requirements are put in place, traceable, and testable from both the technical and user sides.

Feature	Use Case ID	UI ID	Priority	Use Case Cross Ref.	DB Table ID	Elaborated Use Case ID	Dependent Classes
User Login/Registration	UC01	UI01	High	UC02, UC05	DB01	UC01.1, UC01.2	UserAuth, SessionManager
View Academic Records/Percentage	UC02	UI02	High	UC01, UC03, UC05	DB02	UC02.1	RecordsController, Percentage Calculator
Faculty Marks Entry	UC03	UI03	High	UC02, UC04	DB03	UC03.1	MarksManager, GradeValidator

Percentage Calculation Module	UC04	UI04	High	UC02, UC03	DB04	UC04.1	Percentage Calculator, TranscriptService
Transcript Generation	UC05	UI05	Medium	UC01, UC02	DB05	UC05.1	TranscriptService, PDFGenerator
Admin Management Panel	UC06	UI06	Medium	UC01	DB06	UC06.1	AdminPanel, UserManager

Chapter 7: 5th Deliverable (Software Testing)

7.1 Introduction

This chapter introduces the testing plan for the Student Management System with Percentage Calculation, an application used to manage students' records, enter marks, compute Percentage, and print transcripts in a university setting. The application uses role-based access for administrators, faculty, and students. Testing adheres to IEEE Std 829-1998 guidelines to ensure verification of functionality, usability, accuracy, and performance.

The major testing components of this application are:

- Test Plan
- Test Case Specification
- Test Procedure
- Test Summary Report

7.2 Test Plan

7.2.1 Purpose

To establish the scope, methodology, responsibilities, and resources to test the Student Management System with Percentage Calculation to ensure that all the critical features work as expected.

7.2.2 Test Plan ID

System Overview

The system facilitates viewing of academic records and Percentage by the students, entering of marks by the faculties, and managing users, courses, and the system settings by the administrators.

7.2.3 Test Items

- Web-based user interfaces
- Admin, faculty, and student dashboards
- Marks input and Percentage calculation functionality

- Transcript generation module

7.2.4 To Be Tested Features

- Login and registration of all user roles
- Viewing calculated Percentage and academic records
- Marks entry by teaching staff
- Auto-Percentage calculation from grades
- Generation and download of transcripts
- Role-based access control for students, teaching staff, and admin

7.2.5 Features Not Tested

- Offline academic processes not governed via the system
- Overriding Percentage manually by unauthorized staff

7.3 Testing Strategy

The following tiered testing levels will be used

- Unit Testing – Test isolated modules (e.g., Percentage computation, login)
- Integration Testing – Test interaction between system modules
- System Testing – Test the fully assembled system
- User Acceptance Testing (UAT) – Ensure the system satisfies university user requirements

7.3.1 Pass/Fail Criteria

- ✓ Pass: Feature works as anticipated from system requirements
- ✓ Fail: Feature does not work as expected or gives wrong output

7.3.2 Suspension/Resumption Criteria

- ✓ Testing is halted if key features (e.g., login, marks input) fail
- ✓ Testing begins again after critical problems are addressed

7.3.3 Deliverables

- ✓ Test cases documented
- ✓ Bugs/defects reported in detail
- ✓ Thorough test summary report

7.4 Testing Tasks

- Running all test cases
- Tracking and recording bugs in the system
- Regression testing after fixing bugs

7.4.1 Test Environment

- Browsers: Chrome, Firefox
- Devices: Laptop, Desktop
- Database: MySQL (or chosen RDBMS)
- Server: Local or cloud-based web server

7.4.2 Responsibilities

Role	Responsibility
QA Team	Design and execute test cases, report defects
Development Team	Resolve bugs and enhance system stability
Project Admin	Provide final approval for system deployment

7.4.3 Training Needs

All user types (students, instructor, admin) will undergo familiarization training on system features as offered by the QA team.

7.4.4 Schedule

Week	Activity
Week 1	Unit and Integration Testing
Week 2	System Testing
Week 3	User Acceptance Testing (UAT)

7.5 Risks

- ✓ Development delays impacting testing timeline
- ✓ Inadequate availability of system resources or users for UAT
- ✓ Risk of data inconsistencies when test environment is not set up correctly

7.6 Approvals

Testing plan, execution outcomes, and overall system readiness need approval from:

- ✓ QA Lead
- ✓ Development Lead
- ✓ University Administration/Project Stakeholders

Chapter 8: Screenshots of Projects

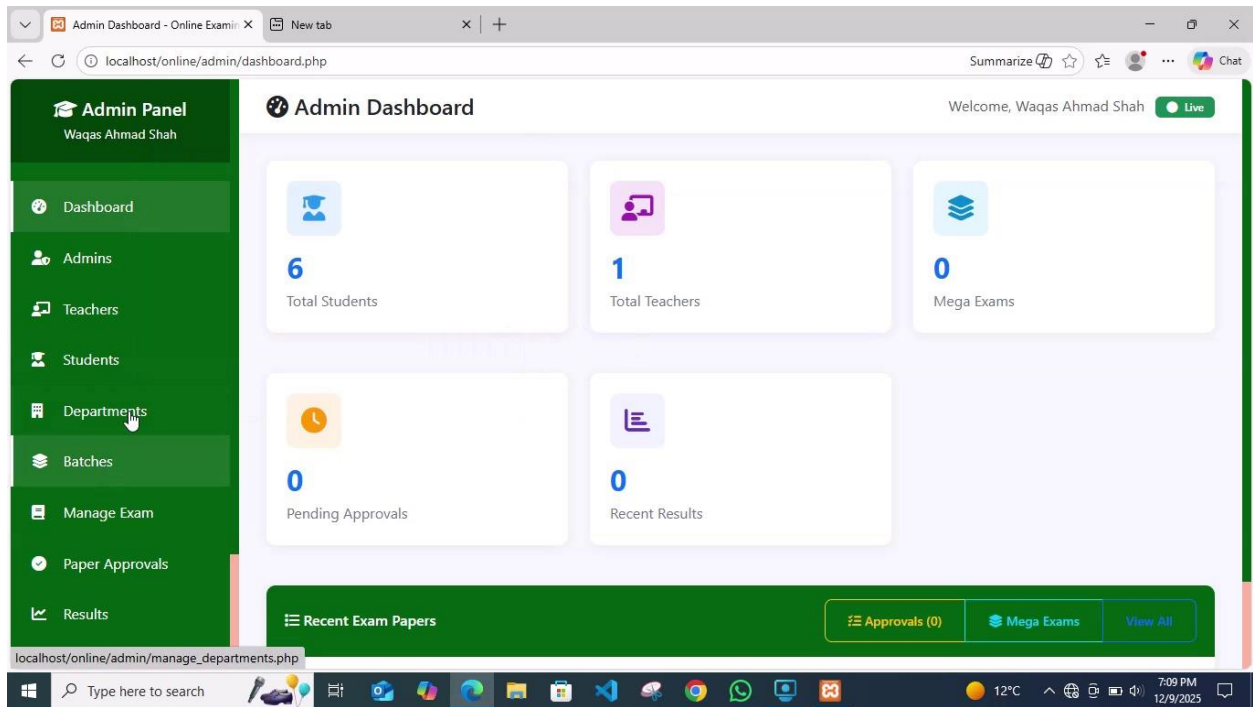
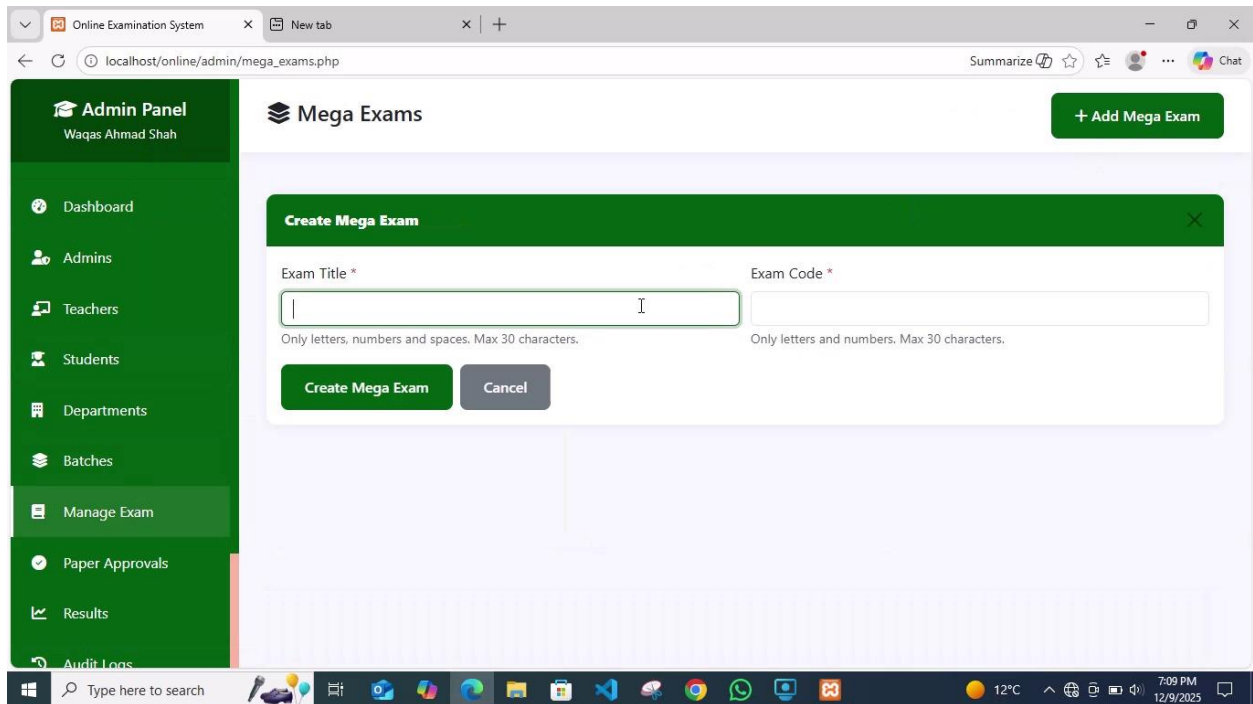


Figure 8. 1 Admin Dashboard



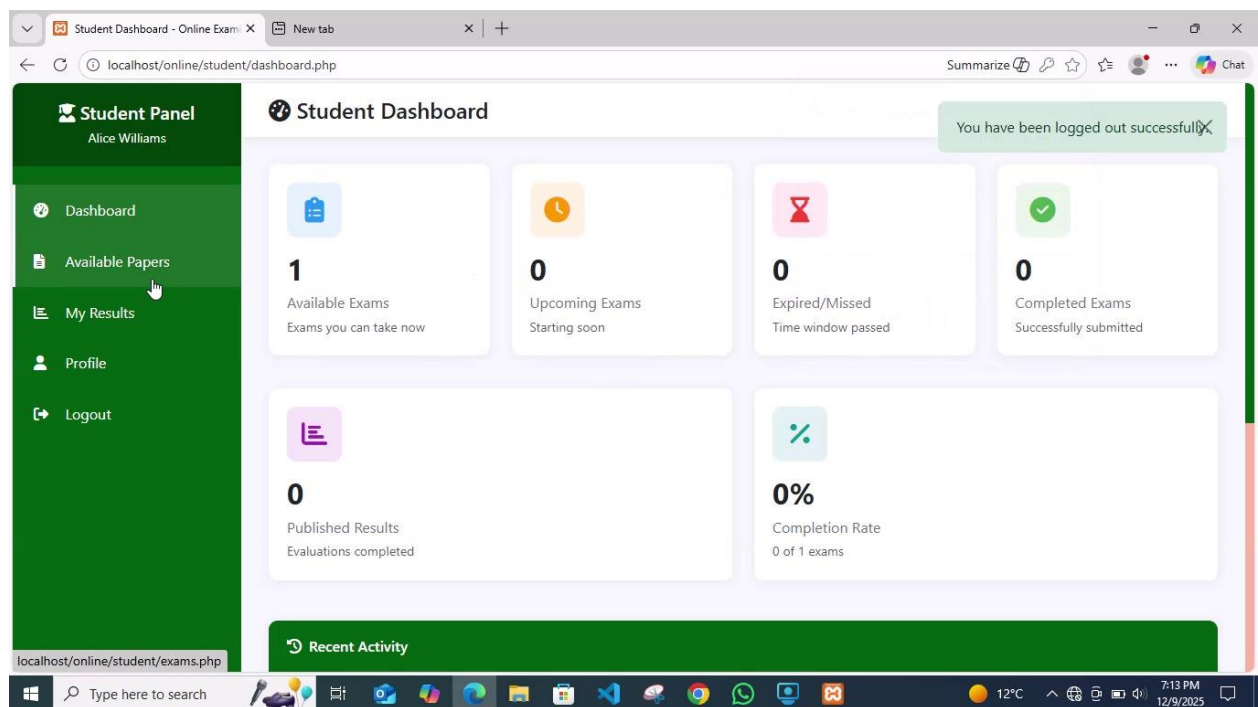


Figure 8. 2 Mega Exam panel

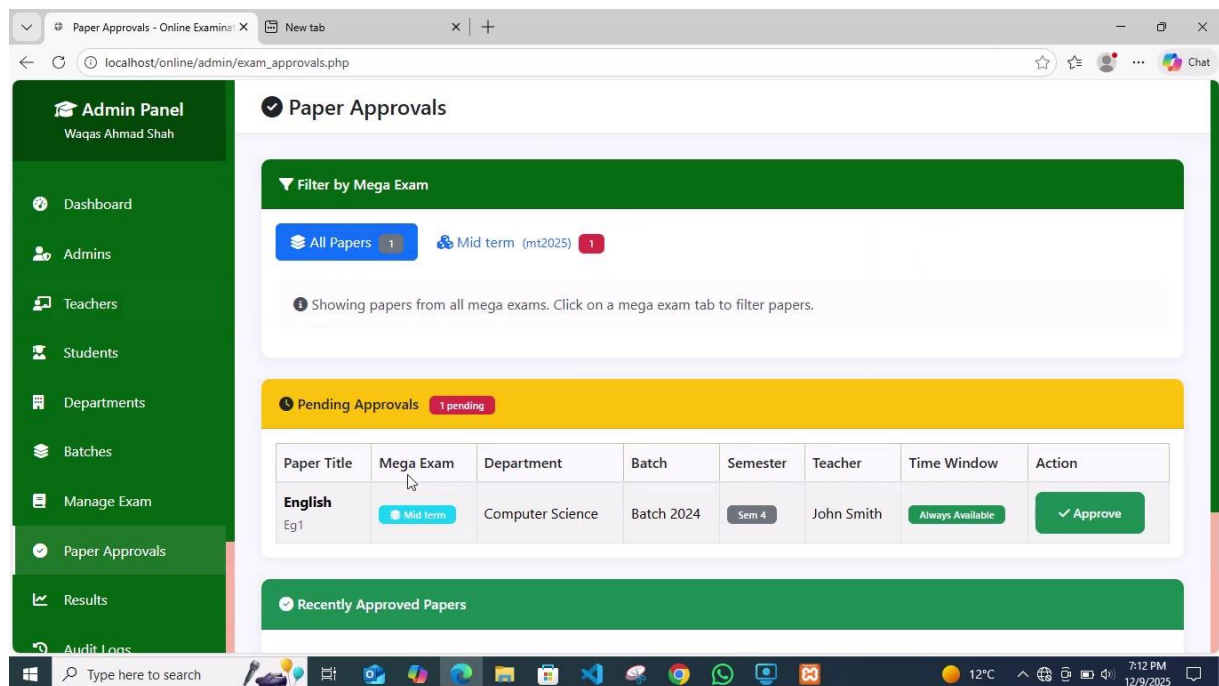


Figure 8. 3 Student Dashboard

Figure 8. 4 Paper Approval

The screenshot shows a web browser window with the URL `localhost/online/student/take_exam.php?exam_id=53`. The page features a green header with the college logo and name, "Govt Degree College Ekkaghund". Below the header, a table displays exam details: Department (Computer Science), Paper (English), Student (Alice Williams), Mega Exam (Mid term), Semester (4), Roll No (CS2024001), Batch (Batch 2024), Duration (20 minutes), and Total Marks (20). A form labeled "Enter Exam Password" includes a text input field and a green "Start Exam" button. The Windows taskbar at the bottom shows the time as 7:13 PM on 12/9/2025.

Govt Degree College Ekkaghund		
Dept: Computer Science	Mega Exam: Mid term	Batch: Batch 2024
Paper: English	Semester: 4	Duration: 20 minutes
Student: Alice Williams	Roll No: CS2024001	Total Marks: 20

Enter Exam Password

Start Exam

Figure 8. 5 Student Login

The screenshot displays the student login interface. The header includes the college logo, name, and "Online Examination System". A green bar shows the time as 04:57. Below, a table lists exam details: Department (Computer Science), Paper (English), Student (Alice Williams), Roll No (CS2024001), Semester (4), and Duration (5 minutes). A yellow button indicates "Current Section: A (mcq)". The main area shows "Question: 1/2" and "Section: A". The question is "Preposition" and the options are "in", "and", "the", and "an". Navigation buttons for "Previous" and "Next" are at the bottom.

Govt Degree College Ekkaghund Online Examination System		
Department: Computer Science	Paper: English	Semester: 4
Student: Alice Williams	Roll No: CS2024001	Duration: 5 minutes

Current Section: A (mcq)

Question: 1/2 Section: A

Question 1 of 2
Preposition

in
and
the
an

Previous Next

Figure 8. 6 Question Display

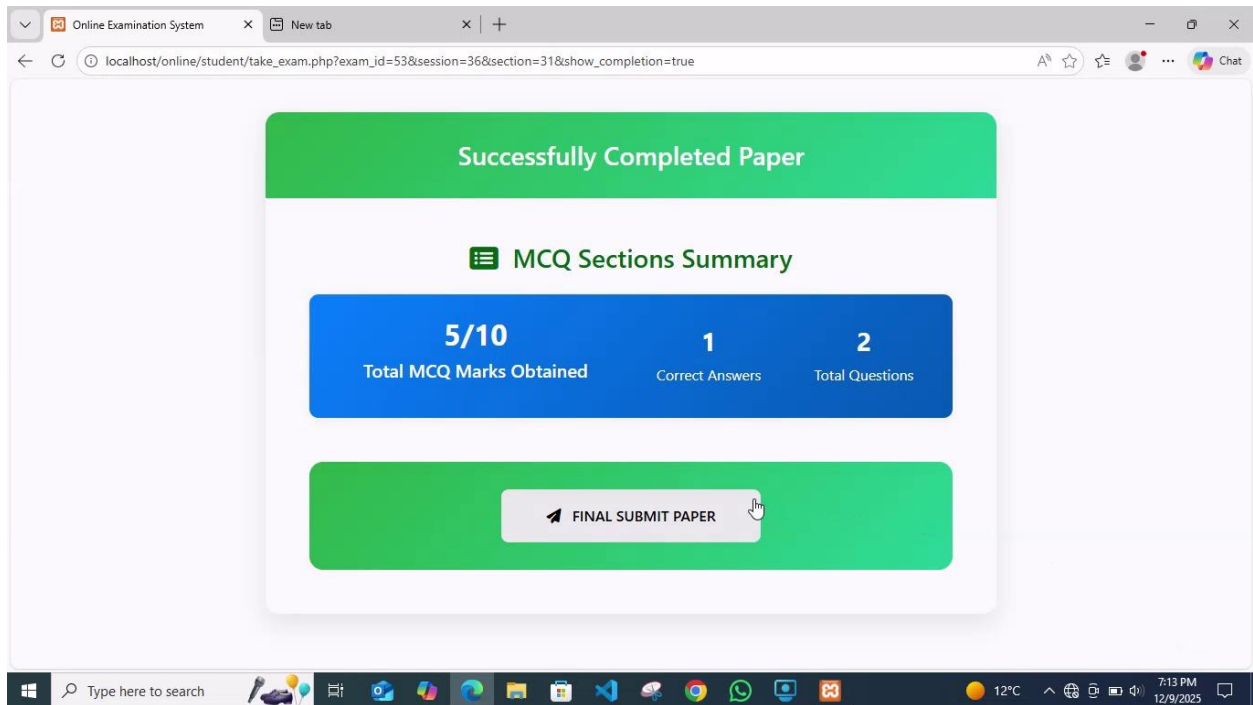


Figure 8. 7 Paper Submission

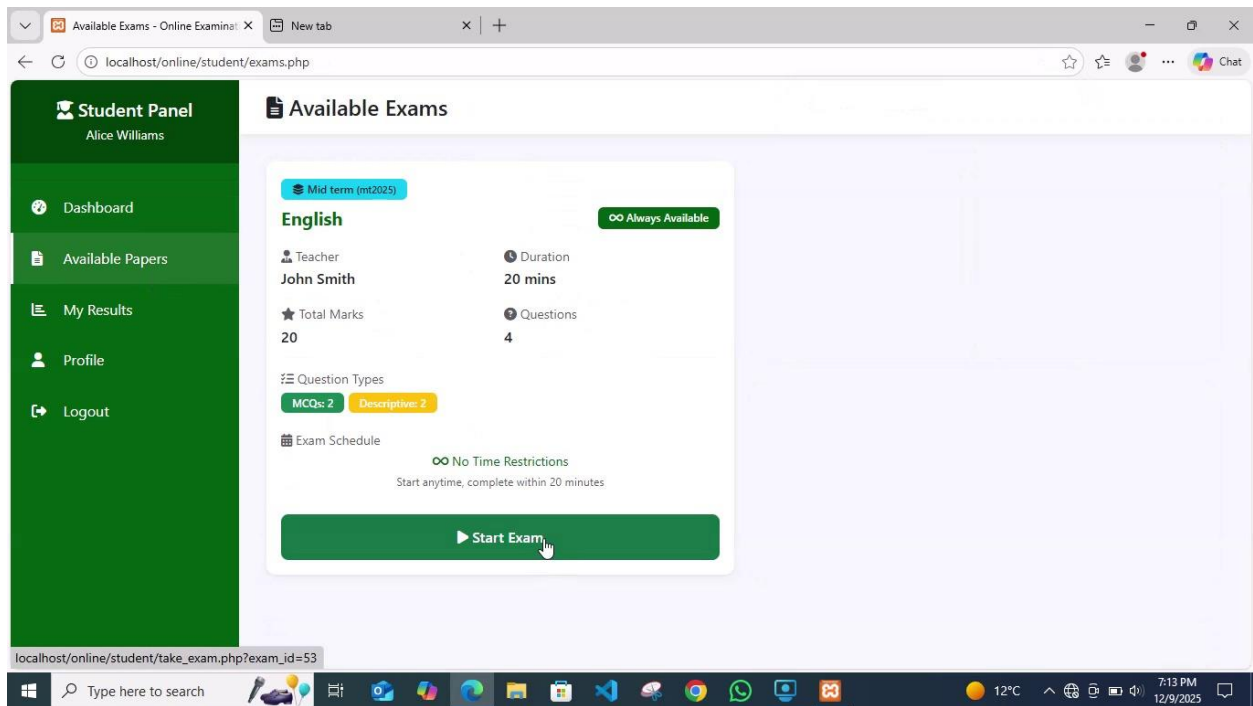


Figure 8. 8 Available Exams Screen

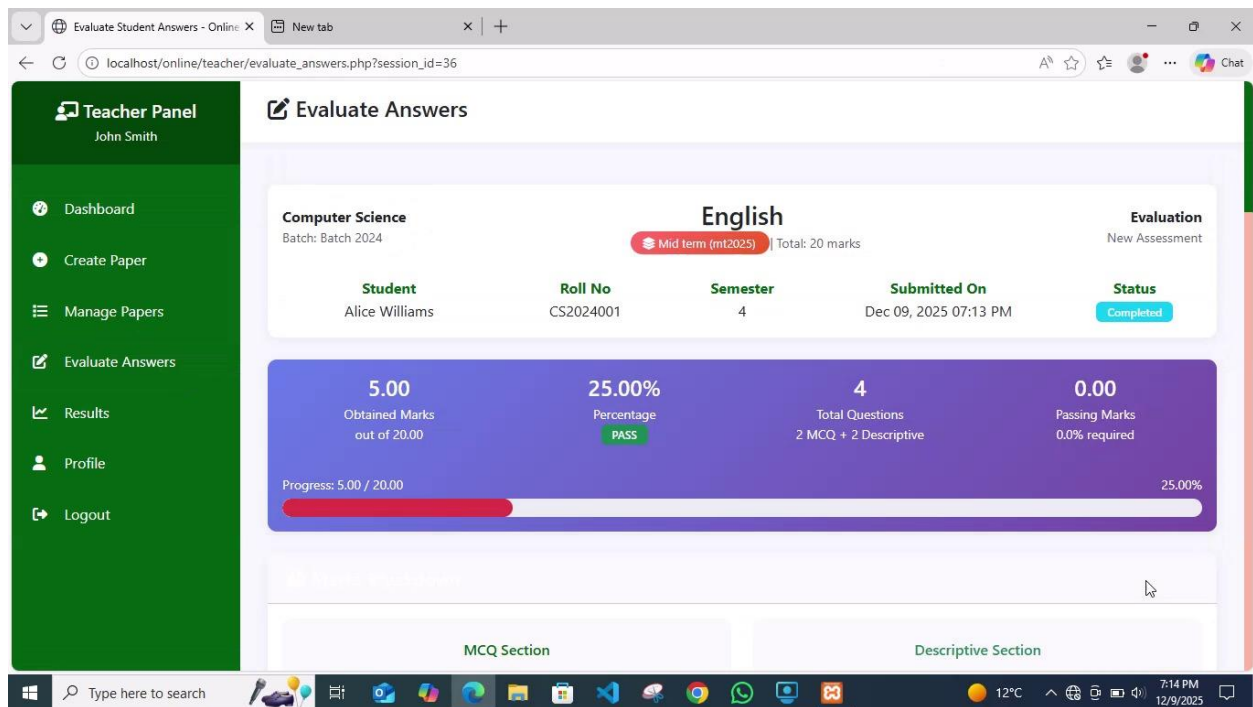


Figure 8. 9 Evaluation Answer

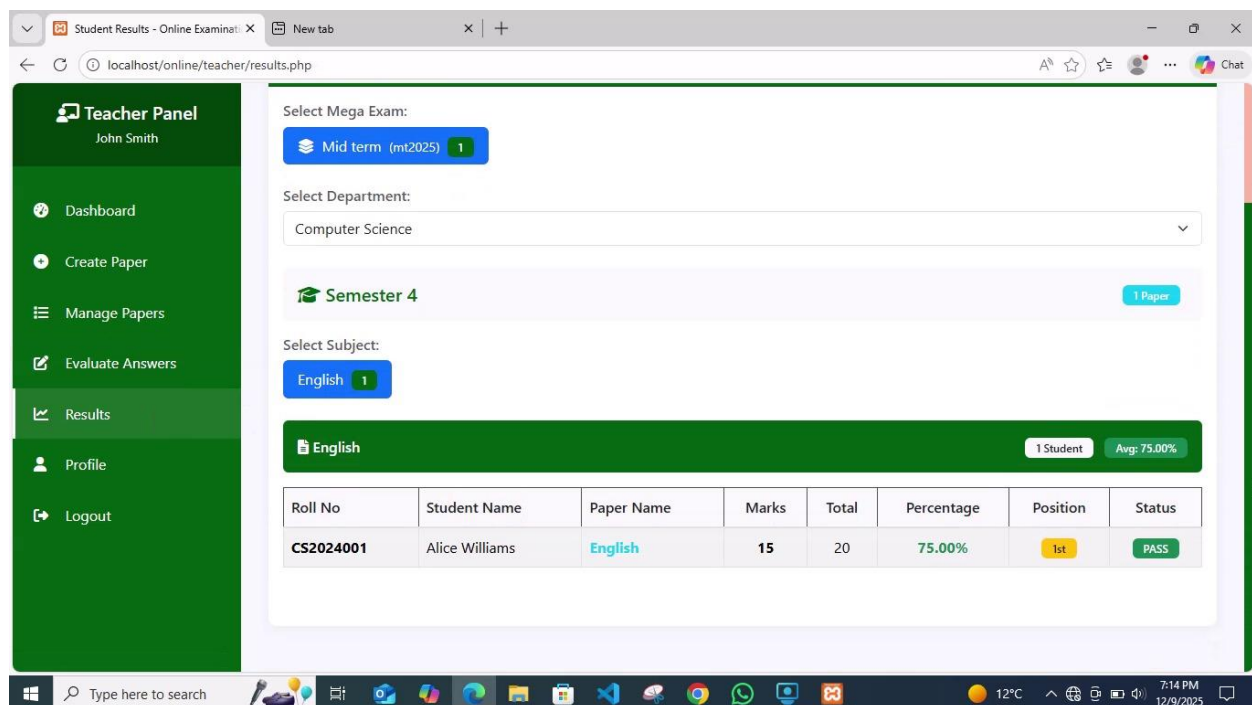


Figure 8. 10 Results Checking

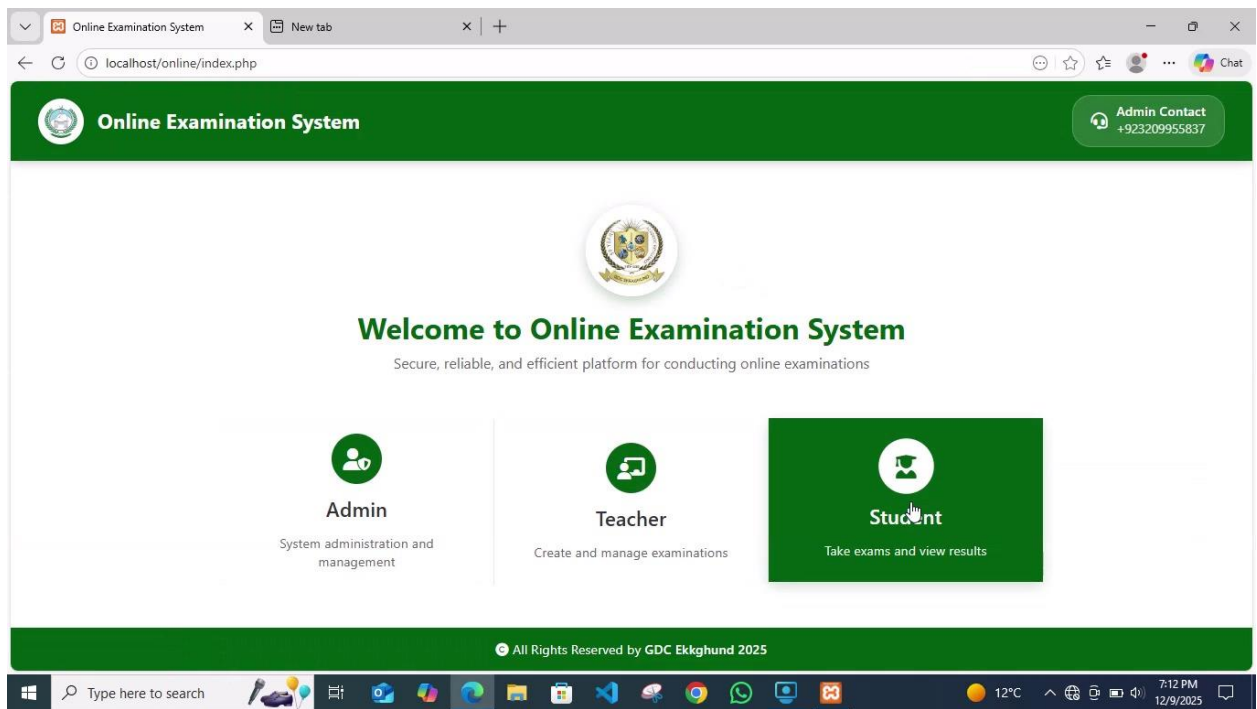


Figure 8. 11 Login Page

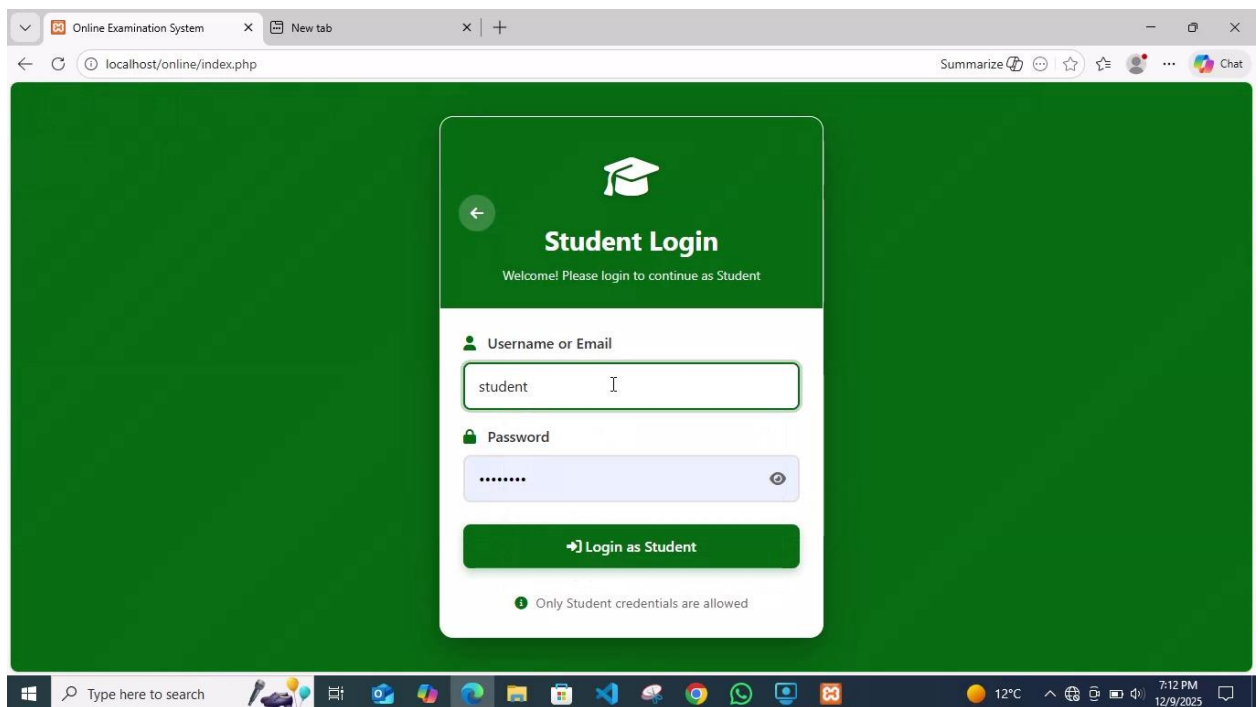


Figure 8. 12 Student Login

Appendixes:

Appendix A – Background Research on:

Microsoft Windows 10 professional/ Window 11

PHP

MySQL

Dreamweaver/ Notepad++

Microsoft Word 2007

Microsoft Visio 2007

Microsoft Paint

Appendix B- Data Dictionary

<http://www.tutorialspoint.com>

<http://www.google.com>

<http://www.php.net>

<https://online.visual-paradigm.com/>

<https://plantuml.online/uml/>